

NVF Case Analysis

Nav D.

December 2024

Table of Contents

Executive Summary	3
Problem Statement	4
Problem Analysis	4
Stakeholders Impacted	5
Constraints	5
Opportunities for Improvement	6
Alternatives	6
AI-Powered Virtual Assistant	6
AI-Human Collaboration Model	9
Comprehensive AI Integration	12
Key Decision Criteria	14
Innovation Level – 40%	15
Implementation Cost – 30%	15
Strategic Alignment with Human-Centered Culture – 30%	16
Alternative Evaluation and Analysis	16
AI-Powered Virtual Assistant	17
AI-Human Collaboration Model	17
Comprehensive AI Integration	18
Recommendation	19
Action and Implementation Plan	20
Conclusion	21
References	22

Executive Summary

This paper provides a detailed analysis of the problem faced by NVF Bank because of growing competition and pressure to adapt to advanced AI technologies. While NVF must maintain its competitive edge, NVF also has to balance technological innovation with its human-centered culture and address concerns about cost and customer satisfaction. Three alternatives identified to address the challenge include:

1. AI-Powered Virtual Assistant
2. AI-Human Collaboration Model
3. Comprehensive AI Integration

These three alternatives are evaluated using a weighted scoring model with key decision criteria – Innovation level, Implementation Cost, and Strategic alignment with Human-Centered Culture. The analysis revealed that AI-powered virtual Assistant is the best option, as it offers meaningful innovation features, cost efficiency, and aligns with NVF's cultural values. Thus, by adopting AI-powered virtual Assistants, NVF would sustain its competitive edge, enhance customer satisfaction, and strengthen its market position.

Problem Statement

NVF could potentially lose its competitive edge to digital native companies if it fails to adapt to technological advancements that align with its overall strategic goal and address operational challenges.

Problem Analysis

In the past NVF Bank has positioned itself as a leading innovator. The bank earned global banking awards and led in digital banking during COVID. However, as the competitors rapidly advance, NVF could lose its competitive edge if it does not adopt advanced technology. The technology team is pushing for aggressive AI adoption to position the organization as a leader in banking innovation. However, this ambition conflicts with NVF's human-centered culture.

Siti, the CEO presented a document to the CFO outlining proposed reductions in workforce and branch operations and handling 95% of customer interactions digitally. While these measures are intended to reduce costs and streamline operations, there are concerns about customer satisfaction. The AI plan outlined by Michael and Sarah involves a \$3 billion investment over five years, with ambitious targets but uncertain return on investment. This creates significant pressure on the organization to deliver measurable outcomes.

NVF's experience with IBM for wealth management services failed to deliver value, so the organization is cautious about significant investments in AI without clearly aligning with NVF's business goals. During a recent Q&A session, students expressed interest in generative AI, reflecting expectations of future talent.

Stakeholders Impacted

This challenge affects stakeholders across NVF. The leadership team is concerned if NVF fails to adopt technological advancements, it risks falling behind competitors, which would harm its reputation and strategic positioning. An aggressive AI strategy could lead to a workforce reduction. Losing experienced employees who deliver NVF's personalized service could weaken the bank's ability to differentiate itself from digital-native competitors, directly affecting its competitive edge. Fully automating processes may isolate customers who prefer human interactions. If NVF does not move quickly enough or fails to integrate AI successfully, the technology team's efforts to establish NVF as a leader in banking innovation could be undermined. This could impact their ability to attract and retain top talent.

Constraints

The financial pressure is to ensure that any AI initiatives deliver measurable results while avoiding the risk of past failures, such as the unsuccessful IBM project. The Plan proposed by the Head of Innovation and CTO includes a significant financial commitment, \$3 billion over five years with no guaranteed return on investment represents a significant risk. Secondly, NVF's reputation is also built on personalized, human-centered service. A sudden shift to an AI-driven process could harm customer satisfaction and loyalty.

The leadership team is also divided on how to pursue AI adoption. While the technology team advocates for rapid innovation to remain competitive, other leaders, such as the CFO, emphasize the importance of maintaining NVF's human-centered culture. Internal misalignment hinders decision-making and creates uncertainty about the path forward.

Opportunities for Improvement

By analyzing the shortcomings of the failed IBM initiative, NVF can identify gaps and better align this AI project with business goals. For example, if IBM's initiative failed due to over-ambitious technological scope or inadequate resource allocation, NVF can avoid similar errors. By clear communication of project goals, adopting a phased implementation plan, and maintaining realistic expectations, NVF can maximize the chances of success of the initiatives.

Alternatives

To address the challenge of maintaining a competitive edge in the market, NVF must consider actionable and measurable alternatives that align with its strategic objectives. The following solutions aim to integrate AI-driven advancements while adhering to operational and budgetary constraints. These alternatives assume that as a bank, NVF may not have all the technical infrastructure required to develop advanced AI solutions independently. So, NVF will collaborate with a software-as-a-service (SaaS) vendor that specializes in AI-powered solutions. NVF's AI team will work with the SaaS vendor to explore innovative approaches in each alternative.

AI-Powered Virtual Assistant

This approach relies on the assumption that NVF has not yet introduced this innovative banking product. While the case study only highlights that NVF has back-office AI and digital projects on fraudulent financial activity and ATM cashouts, customer-facing automation remains an untapped opportunity (Davenport & Westerman, 2024). This approach focuses on deploying a virtual banking assistant such as an AI chatbot to handle routine customer service inquiries and transactional requests. The virtual assistant would function as an AI-powered extension of NVF's

customer service. By automating tasks, NVF can reduce the workload on bank tellers, customer service, and call center representatives, allowing them to focus on more complex or high-value conversations (Zhu, 2022).

This Chatbot would operate as an independent 24/7 system, accessible in any language through NVF's digital channels such as the mobile app and website. It would not enhance human advisors but would directly handle selected customer queries. It would include functionalities such as responding to common customer questions, account inquiries, and money transfers. Research shows that virtual banking assistants can handle up to 80% of customers' queries by responding instantly, improving response time and reducing customer frustrations (Zhu, 2022). The functionalities do not extend to complex banking functions such as financial planning or wealth management.

This alternative focuses on using a generative AI chatbot to stay competitive and keep the digital experience for customers up to date. It will also significantly reduce NVF's operating expenses associated with human support staff. The customer success team and AI team will identify the most common customer scenarios in the research phase to forward to the vendor to create the chatbot. The table below shows the approximate salary cost (Indeed, n.d.). The salaries are from the current payroll budget.

Table 1*Approximate salary cost*

Role	Number of Staff	Approximate Salary	Total Salary
Customer Success Specialist	2	\$55,000	\$110,000
Customer Success Manager	1	\$80,000	\$80,000
AI Project Manager	1	\$90,000	\$90,000
Data Scientist	2	\$100,000	\$200,000

The total yearly estimated salary is \$480,000

The implementation begins with a partnership with a Software-as-a-service (SaaS) vendor that can install pre-trained chatbots for banking workflows (Zhu, 2022). As the SaaS project complexity is basic, the approximate cost will be \$50,000 (Cropolium, 2024). This AI-powered chatbot can be deployed in approximately four weeks (Zhu, 2022). The table below shows an estimated project timeline (Chan, 2020).

Table 2

Project Schedule

Timeline	Activity
4 weeks	Identify Chatbot use cases
2 weeks	Prioritize use cases
4 weeks	Scenario selection
2 weeks	Creating a final scenario list
4 weeks	SaaS-Vendor creates a pre-defined chatbot
4 weeks	Implementation
Starting 5 th month and ongoing	Technical Support by the vendor

An estimate of total employee salaries = Total resource cost x Total time (Terpilowski, 2021)

$$= \$480,000 \times 5/12 \text{ (months)} = \$200,000$$
$$\text{Total project cost} = \text{Total salaries} + \text{SaaS cost} = \$250,000$$

AI-Human Collaboration Model

This approach blends AI with human advisors to enhance customer relationships and satisfaction. AI tools would assist human advisors, call center representatives, and analysts by providing real-time insights, enabling NVF to provide personalized service while maintaining a human-centered culture. The idea is to allow humans and AI to work together with AI handling data-intensive tasks and humans focusing on relationship-building and strategic decision-making (Abramov, 2024).

AI’s functionalities include analyzing customer financial behavior and credit data to provide real-time updates on credit scoring and offer insights on improving creditworthiness. Human advisors can also use this information to have meaningful conversations with customers, offering personalized advice tailored to their financial goals. Similarly, it will streamline loan applications and insurance claims processes. AI algorithms would pre-assess clients' eligibility, and human advisors can step in to discuss complex cases, ensuring a smooth and satisfying customer experience (Abramov, 2024).

The Project management team will identify all the areas where this AI tool can be used and create a list of requirements. The requirements will then be provided to Software-as-a-service (SaaS) vendors, who will customize these AI-powered analytical tools based on the requirements (Zhu, 2022).

The table below shows the approximate salary cost (Indeed, n.d.). The salaries are from the current payroll budget.

Table 3

Approximate salary cost

Role	Number of Staff	Approximate Salary	Total Salary
Business Analyst	4	\$85,000	\$340,000
AI Project Manager	1	\$90,000	\$90,000
Training Specialist	4	\$60,000	\$240,000

The total yearly estimated salary is \$670,000

Lastly, these AI-powered analytics tools will be integrated into NVF’s existing customer service platforms, and employees will be trained to use them effectively. This tool can be used solely by the customer or by human advisors during meetings. As the SaaS project complexity is average, the approximate cost will be \$120,000 (Cropolium, 2024). By offering a hybrid model, these alternatives aim to enhance customer satisfaction. The table below shows an estimated project timeline (Evangelist, 2025).

Table 4

Project Schedule

Timeline	Activity
4 weeks	Develop AI strategy – Identify gaps
2 weeks	Identify highest-value AI opportunities
2 weeks	Create a requirement list for AI analytics tools
4 weeks	SaaS-Vendor build and customize the tools
4 weeks	Implementation
4 weeks	Training
Starting 5 th month and ongoing	Technical Support by the vendor

An estimate of total employee salaries = Total resource cost x Total time (Terpilowski, 2021)

= \$674,000 x 6/12 (months) = \$335,000

Total project cost = Total salaries + SaaS cost = \$455,000

Comprehensive AI Integration

In this approach, NVF would transform into a fully AI-powered organization, integrating artificial intelligence across all operations, including customer-facing services, internal processes, and decision-making frameworks. This approach aims to achieve end-to-end automation and create an autonomous financial ecosystem that optimizes efficiency, reduces cost, and delivers a personalized banking experience with minimal manual intervention (Aurionpro, n.d.).

The transition would be carried out in carefully planned phases to ensure operational efficiency, scalability, and effective implementation. The first phase focuses on automating routine and repetitive tasks that consume significant time and resources, such as data entry and compliance checks. The next phase is to deploy AI analytics tools into NVF's core operations to support smart decision-making. For example, AI algorithms can drive smarter credit scoring and loan underwriting processes by analyzing vast datasets. AI-driven chatbots to handle customer inquiries will also be launched in this phase.

The next phase would be to deploy and use AI models for predictive insights. Lastly, a fully autonomous finance system would allow the organization to operate autonomously with minimal human intervention. The AI system would manage all operations, from onboarding to regulatory compliance and financial advisory. By fully embracing AI, NVF will position itself as autonomous finance, creating a significant competitive advantage over digital-native competitors (Aurionpro, n.d.).

The implementation requires expertise in AI model development, infrastructure upgrades, and data sciences (First Line Software, n.d.). The table below shows the approximate salary cost (Indeed, n.d.). The salaries are from the current payroll budget.

Table 5

Approximate salary cost

Role	Number of Staff	Approximate Salary	Total Salary
AI Project Manager	1	\$90,000	\$90,000
Data Scientist	2	\$100,000	\$200,000
Infrastructure Engineer	2	\$120,000	\$240,000

The total yearly estimated salary is \$530,000

As the SaaS project complexity is complex, the approximate cost is \$300,000 (Cropolium, 2024). The table below shows an estimated project timeline (Evangelist, 2025).

Table 6

Project Schedule

Timeline	Activity
8 weeks	Audit current processes and identify the gap
8 weeks	Phase I: SaaS-Vendor builds and deploys AI models for core processes
8 weeks	Phase II: Vendor integrates AI analytics
8 weeks	Phase III: Launch predictive analytics models

8 weeks	Phase IV: Full-scale testing and deployment across all operations
Starting 11 th month and ongoing	Monitor performance and conduct ongoing optimization

An estimate of total employee salaries = Total resource cost x Total time (Terpilowski, 2021)

$$= \$530,000 \times 10/12 \text{ (months)} = \$441,667$$

$$\text{Total project cost} = \text{Total salaries} + \text{SaaS cost} = \$741,667$$

Key Decision Criteria

Key decision criteria are the foundation for evaluating alternatives and selecting the best option. These criteria address the issues and opportunities identified in the problem statement and ensure that the decision-making process aligns with NVF’s goals (Breakcold, 2024). The key decision criteria to evaluate alternatives in the NVF case are innovation level, implementation cost, and strategic alignment with human-centered culture. These criteria are measurable, directly tied to the problem, and allow for a clear comparison of alternatives (Breakcold, 2024).

NVF must add technological advancements to maintain a competitive edge in the market, so the innovation level is the most important factor and is given a weight of 40%. Implementation cost and strategic alignment with human-centered culture are given a weight of 30% acknowledging their importance in decision-making.

Innovation Level – 40%

Innovation is critical to maintaining NVF's competitive edge in the financial sector. The innovative level measures how much each alternative introduces impactful and meaningful technological advancements that ensure NVF stays ahead of the curve. It showcases an organization's ability to convert investments into innovations. Alternative that is revolutionary will score high on the scale (Terwilliger, 2015).

- High innovation (Transformational innovation): An alternative that offers cutting-edge, AI-powered services will be given 3 points.
- Moderate innovation (Breakthrough innovation): An alternative that introduces impactful but gradual technological advancement will be given 2 points.
- Low innovation (Incremental innovation): An alternative that focuses on small, meaningful improvements will be given 1 point.

Implementation Cost – 30%

Cost efficiency is also one of the key factors in NVF's decision-making process to solve the current issue. An ideal alternative would be financially feasible, and address NVF operational challenges, thus balancing innovation with cost-effectiveness. Each alternative's total estimated implementation cost will be compared and an alternative with a streamlined setup and minimal cost will be given higher points.

- High cost: An alternative that will cost more than \$500,00 will be 1 point.
- Moderate cost: An alternative costing more than \$250,00, less than \$500,00 will be given 2 points.
- Low cost: An alternative costing less than \$250,00 will be given 3 points.

Strategic Alignment with Human-Centered Culture – 30%

One pillar of NVF's reputation is personalized, and human-centered services (Davenport & Westerman, 2024). As the organization integrates AI, it is critical to ensure that the selected alternative enhances customer satisfaction and upholds NVF's cultural values. Strategic alignment would ensure that technological advancement fosters customer loyalty and trust (Cascade, 2023). This criterion assesses how well each alternative will ensure NVF retains its competitive edge without compromising its identity.

- Strong alignment: An alternative that fully preserves NVF's human-centered culture while leveraging AI to enhance customer experience will be given 3 points.
- Partial alignment: An alternative that balances AI tools with human services maintaining some level of customer interaction will be given 2 points.
- Poor alignment: An alternative that completely excludes NVF's human-centered values will be given 1 point.

Alternative Evaluation and Analysis

The three proposed alternatives Efficient AI-powered customer support, Collaborative AI-human customer engagement, and comprehensive AI Integration across the organization are analyzed against key decision criteria. The weighted scoring model is used for the evaluation of all alternatives. Each alternative is scored against the key decision criteria to determine which alternative addresses NVF's challenges and opportunities. To calculate the total score and rank alternatives, the score received by each alternative for a decision criterion is multiplied by the percentage weight of the decision criterion (Faisal, n.d.).

AI-Powered Virtual Assistant

This approach offers a modern solution by automating routine tasks such as customer inquiries, account management, and money transfers. The virtual assistant (chatbot) can chat in any language, thus can help NVF's diverse customers. While this solution is not groundbreaking but involves AI to enhance customer service accessibility and efficiency. As the innovation is breakthrough, not transformational this alternative scores 2 points for the innovation level.

The estimated total cost for SaaS implementation and staff salaries is \$250,000 which makes this a low-cost alternative. So, this alternative scored 3 points for the implementation cost criterion. Lastly, the virtual assistant prioritizes convenience for customers while maintaining service quality by providing instant, 24/7 access to simple banking services. It focuses on enhancing the customer experience without compromising NVF's human-centered values, so it scores 3 points on the strategic alignment criterion.

$$\text{Total score} = (2 \times 0.4) + (3 \times 0.3) + (3 \times 0.3) = 2.6$$

AI-Human Collaboration Model

This alternative introduces AI tools to assist human advisors and enhance customer interactions. As the innovation focuses on credit scoring and streamlining loan applications, it does not fully transform NVF's operations, so it scores 2 points on the innovation level criterion. With a total estimated cost of \$455,000, this alternative is moderately expensive. So, this alternative scores 2 points for the implementation cost criterion.

As this alternative balances automation with human involvement, it preserves NVF's human-centered culture. AI supports customers and employees in decision-making thus

enhancing customer relationships and maintaining trust. So, this alternative scores 3 points as it demonstrates strong alignment with NVF's cultural values.

$$\text{Total score} = (2 \times 0.4) + (2 \times 0.3) + (3 \times 0.3) = 2.3$$

Comprehensive AI Integration

This alternative represents a transformational shift to a fully AI-powered organization, by automating core processes and enabling predictive and proactive financial management. By redefining NVF's operational model this alternative has the potential to disrupt the industry and create new standards. So, it scores 3 points on the innovation level criterion. The total estimated implementation cost is \$741,667 which makes this an expensive alternative, so it scores 1 point for cost criterion.

As this alternative focuses on transforming NVF into a fully autonomous organization by deploying AI across all operations it risks NVF's human-centered culture. The emphasis on full automation with minimal human involvement could isolate customers who value personal interaction, so this alternative scored 1 point for the strategic alignment criterion.

$$\text{Total score} = (3 \times 0.4) + (1 \times 0.3) + (1 \times 0.3) = 1.8$$

Table 7 is the summary table listing all alternatives, key decision criteria, and the score received by each alternative. The total Score reflects the weight.

Table 7

Summary of evaluation of alternatives against key decision criteria.

Alternatives	Innovation level	Implementation cost	Strategic alignment	Total Score 100%
	40%	30%	30%	
AI-Powered Virtual Assistant	2	3	3	2.6
AI-Human Collaboration Model	2	2	3	2.3
Comprehensive AI Integration	3	1	1	1.8

Recommendation

Based on the evaluation of the alternatives against the key decision criteria, the recommended alternative is the AI-powered virtual assistant alternative. This alternative scored the highest overall score of 2.6, effectively addressing NVF's challenges while aligning with its strategic goals. While this alternative introduces only breakthrough innovation, it effectively modernizes customer service by automating routine tasks and providing customer service in any language. These meaningful straightforward advancements help NVF maintain its competitive edge without putting much at stake. This alternative also aligns well with NVF's operational scale making it a financially feasible and impactful choice.

This AI-powered virtual assistant also preserves NVF's human-centered culture by enhancing customer experience through 24/7 accessibility, reducing wait and response times. It does not replace human involvement entirely, customers who value in-person services are still supported.

Action and Implementation Plan

A clear and structured approach is necessary to ensure the successful adoption of AI-powered virtual assistant initiatives. The action and implementation plan is a step-by-step guide to turning strategies into tasks (Asana, 2025). By adhering to this phased approach, NVF can mitigate risk, manage cost-effectively, and ensure alignment with its human-centered culture. Leveraging this technological advancement, NVF can sustain its competitive edge in the market. The plan outlined in Table 8 outlines the specific steps, actions, schedule, and stakeholders' information.

Table 8

Action and Implementation Plan

Steps	Tasks	Stakeholders Involved	Duration	Start date	End date
Research	Clearly identify the purpose of chatbot Identify all Chatbot use cases	Customer Success Specialist, and Customer Success Manager	4 weeks	1/27/2025	2/24/2025
Selection - Phase I	Create list of all chatbot use cases Prioritize use cases	Customer Success Specialist, Customer Success Manager and AI team	2 weeks	2/25/2025	3/11/2025
Selection - Phase II	Select scenarios for the chatbot Select a SaaS vendor	AI team and Customer Success Team AI Project Manager and Data Scientist	4 weeks	3/12/2025	4/9/2025
Requirements	Document all the requirements for the product	AI team and Customer Success Team	2 weeks	4/10/2025	4/25/2025
Development	Develop Prototype Feedback Refinement Feedback Testing	SaaS vendor AI Project Manager and Data Scientist SaaS Vendor AI Project Manager and Data Scientist SaaS Vendor and AI team	4 weeks	4/28/2025	5/12/2025
Implementation	Integration of chatbot to NVF's mobile app and website	SaaS vendor and AI team	4 weeks	5/13/2025	6/10/2025
Post-Implementation	Technical Support Feedback collection	SaaS vendor and AI team AI team, Customer Support team, and SaaS vendor	Ongoing		

Note: This is an estimated timeline and schedule only

Conclusion

NVF Bank stands at a critical point where it must adopt innovative technology to retain its competitive position in the market. Thorough analysis revealed that the AI-powered virtual assistant alternative emerges as the optimal solution to address NVF's challenges and opportunities. This alternative balances the need for breakthrough innovation, cost efficiency, and alignment with the bank's human-centered culture. This AI-driven alternative offers a sustainable path, that ensures that the bank remains a leader in innovation and customer satisfaction.

References

- Abramov, M. (2024, August 2). *The Human-AI collaboration: How humans and AI can work together in finance*. <https://keymakr.com/blog/the-human-ai-collaboration-how-humans-and-ai-can-work-together-in-finance/>
- Asana. (2025, January 6). *What is an implementation plan? 6 steps to create one*.
<https://asana.com/resources/implementation-plan>
- Aurionpro. (n.d.). *The Evolution of Autonomous Finance: Moving toward fully AI-driven banking*. <https://www.aurionpro.com/blogs/the-evolution-of-autonomous-finance-moving-toward-fully-ai-driven-banking/>
- Breakcold. (2024, September 11). *What are Decision criteria? (Explained with examples)*.
<https://www.breakcold.com/explain/decision-criteria>
- Cascade. (n.d.). *Strategic alignment: why it matters and how you do it*.
<https://www.cascade.app/blog/strategic-alignment-why-it-matters>
- Chan, D. (2020, November 9). *Chatbot Use Cases: A How-To guide*.
<https://www.purespeechtechnology.com/conversational-ai/chatbot-use-cases-a-how-to-guide/>
- Cropolium. (2024, April 19). *How much does it cost to build a SaaS app: Breakdown*.
<https://acropolium.com/blog/how-much-does-it-cost-to-build-a-saas-app-in-2022/>
- Davenport, T. & Westerman, G. (2024). *How Aggressively should a bank pursue AI?*
<https://hbr.org/2024/05/case-study-how-aggressively-should-a-bank-pursue-ai>

Faisal, S. (n.d.). *Weighted Scoring Model: What it is & how to create it.*

<https://userpilot.com/blog/weighted-scoring-model/>

First Line Software. (n.d.). *10 Steps to consider adopting AI in your business.*

<https://firstlinesoftware.com/blog/ten-steps-to-consider-adopting-ai-in-your-company/>

Indeed. (n.d.) *Discover your earning potential.* <https://www.indeed.com/career/salaries>

Kardaszynska, D. (2024, March 20). *The Rise of virtual assistants: AI's impact on banking*

customer care. <https://livebank24.com/artificial-intelligence/the-rise-of-virtual-assistants-ais-impact-on-banking-customer-care/>

Terpilowski, A. (2021, March 16). *How do you calculate overhead cost per employee?*

<https://www.bigtime.net/blogs/overhead-per-employee/>

Terwilliger, J. (2015, September 30). *The three levels of Innovation.*

<https://www.creativerealities.com/innovationist-blog/bid/49954/the-three-levels-of-innovation>

Zhu, S. (2022, March 24). *5 Reasons Every Bank must a virtual banking assistant.*

<https://thefinancialbrand.com/news/banking-technology/5-reasons-every-bank-must-have-a-virtual-banking-assistant-131841>